MILITARY SPECIFICATION

CARTRIDGE, 105MM, TANK GUN FLASH SIMULATOR, XM23

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the parts, loading, assembling and packing of 105MM Tank Gun Flash Simulator.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC:1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGE DRAWINGS

9341575 - 105MM, Tank Gun Flash Simulator

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable specifications and drawings.

3.1.1 Usage of materials. All materials and parts used in assembling 105MM Tank Gun Flash Simulator cartridges shall be used on a first-in, first-out basis. This requires that the materials and parts received by the loading plant will be utilized in a sequential or chronological order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

3.2 Components and complete rounds. The components and completed rounds shall comply with all requirements specified on drawing (dwg.) 9341575 and with all requirements specified in applicable specifications, standards and drawings.

3.3 Cartridge functioning. The 105MM Tank Gun Flash Simulator shall function in such a way that it shall produce audible evidence of the sound and visible evidence of the flash produced by a normal tank round.

3.4 Safety. Cartridges shall comply with the following safety criteria.

3.4.1 Misfires. The cartridges shall not fail to fire from the weapon.

3.4.2 Hangfires. The cartridges shall have no perceptible delay between the instant of application of energy and functioning of the cartridge.

3.5 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the below listed paragraphs.

3.5.1 General. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. The cartridges shall not be damaged in any manner during the loading, assembling and packing operations.

3.5.2 Marking. The marking shall be restored by the assembling contractor on any components furnished to him marked, when such markings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)

b. Quality conformance inspection.(see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of paragraph 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Bag, Charge	9341576	10
Charge	9341578	10
Disc, Cardboard	9342936	5
Disc, Cork	9342935	5
Ring, Split	9343964	10
105MM Tank Gun Flash Simulator	9341575	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

TABLE I FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, 105MM, Tank Gun Flash Simulator				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bag, Charge (Dwg. #9341576) Examination for defects	10		3.2	4.4.2.1
	Charge (Dwg. #9341578) Examination for defects	10		3.2	4.4.2.2
	Disc Cardboard (Dwg. #9342936) Examination for defects	5		3.2	4.4.2.3
	Disc, Cork (Dwg. #9342935) Examination for defects	5		3.2	4.4.2.4
	Ring Split (Dwg. #9343964) Examination for defects	10		3.2	4.4.2.5

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

TABLE I FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, 105MM, Tank Gun Flash Simulator				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	105MM, Tank Gun Flash Simulator (Dwg. #9341575)				
	Examination for defects	10		3.2	4.4.2.6
	Cartridge functioning	10		3.3	4.4.3.1

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition inspection lots of 105MM Tank Gun Flash Simulator Cartridges shall contain:

a. Primers of one lot interfix number from one manufacturer.

b. Cartridge cases of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, General Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Bag, Charge				9341576
					NEXT HIGHER ASSEMBLY
					9341578
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag, charge, cloth torn or punctured		100%	3.2	Visual
102	Stitching missing or broken		100%	3.2	Visual
Minor					
201	Unravelling of cut edges		0.65%	3.2	Visual
202	Back tack missing from the ends of the longitudinal seams		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Charge		9341578
			NEXT HIGHER ASSEMBLY
			9341575

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant weight		100%	3.2	4.5.1
102	Top, sealing seam missing or broken		100%	3.2	Visual
Minor					
201	Marking missing, misleading, marred or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Disc, Cork				9342935
					NEXT HIGHER ASSEMBLY 9341575
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Disc diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Disc, Cork				9342935
					NEXT HIGHER ASSEMBLY
					9341575
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Disc diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Ring, Split				9343964
					NEXT HIGHER ASSEMBLY
					9341575
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Wall thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63526(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	105MM Tank Gun Flash Simulator				9341575
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Electrical resistance		100%	3.2	4.5.2
102	Primer above flush		100%	3.2	4.5.3
103	Primer disassembly torque		0.40%	3.2	4.5.4
104	Profile, alignment and chambering		100%	3.2	4.5.5
105	Excessive depth to primer electrode		0.40%	3.2	Gage
106	Epoxy depth		0.40%	3.2	Gage
107	Carboard disc, cork disc and split ring properly installed		0.40%	3.2	Visual
108	Foreign matter in case		0.40%	3.2	Visual
Minor					
201	Primer not staked		0.65%	3.2	Visual
202	Primer bent or damaged		0.65%	3.2	Visual
203	Markings missing, misleading, marred or unidentifiable		0.65%	3.5.2	Visual
204	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described below and in the Quality Conformance Inspection Tables.

4.4.3.1 Proving ground functional tests. These tests shall be performed at a proving ground in accordance with the applicable Acceptance Test Procedure. All recorded information shall be provided in the firing reports. Noise levels (db) shall be measured at the gun position and recorded. The sample cartridges shall be electrically fired using energy, equivalent to the service tank's firing circuit. Any misfire or hangfire (see 3.4.1, 3.4.2) shall be recorded in the firing record and reported. Firings shall be from a fixed type facility mount or from an M60 series tank mounting a suitable 105MM M68 gun tube. The tube shall be visually inspected for adequacy prior to initiating each firing program. Inspections shall occur more frequently when there are unexpected occurrences such as cartridge case rupture, chamber pressures in excess of 117 percent of upper pressure limit (117% upl = 72,300 psi), etc., during a firing program. In addition, the gun tube will be monitored for any damage, undue wear or any other adverse effects as might be as a result of the firing of the 105MM Tank Gun Flash Simulator Cartridge. Observations shall be made for conformance to all requirements (see 3.3). Retesting shall be conducted as demanded by adverse test conditions (see 6.5).

4.4.3.1.1 Safety. All cartridges fired shall be observed for hangfires and misfires. Any cartridge exhibiting a safety failure shall be cause for an immediate halt to testing of the lot and shall be cause for rejection of the lot.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit, for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078. Submission shall be in accordance with DI-R-1714 with Addendum I.

4.5 Methods of inspection.

4.5.1 Propellant weight. The propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any propellant charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot, subject to correction prior to resubmission.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.5.2 Electrical resistance. The electrical resistance shall be measured between the primer electrode and the cartridge case with a maximum current of 20 milliamperes flowing through the primer. Any case-primer assembly which fails to meet the resistance requirement shall be classed defective and be removed from the lot.

4.5.3 Primer above flush. The cartridge shall be checked for primer above flush with the base of the case. Any cartridge which fails to meet the applicable requirements shall be classed defective

4.5.4 Primer disassembly torque. After the primer has been staked, the minimum disassembly torque shall be applied on the primer with a suitably fixtured torque wrench. The primer must withstand this minimum disassembly torque without moving. If the number of primers that move exceeds the applicable requirement, the lot shall be rejected. Cartridge cases with primers that pass this test may be returned to the lot.

4.5.5 Profile and alignment. The cartridge must freely enter the profile and alignment gage. The cartridge shall be loaded horizontally with 60 pounds maximum force at a rate of 2 feet per second. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The cartridge shall be preserved and packaged in accordance with dwg. 9293480.

5.2 Packing and marking.

5.2.1 Level A. The containers shall be packed in accordance with dwg. 9293478.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

5.3 Marking. The containers and boxes shall be marked in accordance with dwgs. 9293478 and 9293480.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use from the M68 tank gun.

6.2 Ordering data. See MIL-A-48078.

6.2.1 Data requirements. When this specification is used in a procurement which incorporates of DD Form 1423 and invokes the provision of 7-104.9 (n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9 (n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.4	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714

(Copies of data item descriptions required by the contractor in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Check test for possible deterioration of primers. If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at anytime since previous tests, the primer lot shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in the applicable primer specification immediately before the primer lot is assembled into

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

the cartridges. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading ARRADCOM may also include drawings prepared by and identified as Picatinny Arsenal drawings. Technical data originally prepared by this activity is now under the cognizance of ARRADCOM.

6.5 Retest. Additional cartridges, as required shall be tested when the proving ground considers that test conditions have detrimentally affected the test results. All conditions of 4.4.3.1 shall apply for retest provisions. The results of the test cartridges fired during the retest shall not be pooled with the results of the initial test cartridges.

Custodian:
Army-AR

Preparing Activity
Army-AR

Project Number: 1315-A450

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
CARTRIDGE, 105MM, TANK GUN FLASH SIMULATOR, XM23 MIL-C-63526

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT COMMAND
ATTN: DRDAR-QAR
DOVER, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.